Windows Registry Monitoring System

Monitoring Results & Testing Documentation

1. Baseline Creation

Command Executed:

- python create_baseline.py

This command creates an initial snapshot (baseline) of all configured registry keys and stores them inside baseline/baseline_registry.json. This snapshot is later used for comparison.

2. Start Monitoring Engine

Command Executed:

- python main.py

This starts the continuous polling engine. The system reads configured registry paths every 5 seconds and compares them with the baseline to detect changes.

3. Manual Registry Modification (Testing Step)

To test detection capability, a new registry value was manually added in:
HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Run

Test Value Name: test_realtime_check
Test Value Data: C:\Users\Public\test.exe

4. Monitoring Output Detection

After modifying the registry, the monitoring engine detected the following change types:
- ADDED
- MODIFIED
- DELETED

Below are the real-time console outputs captured during testing:

```
('DELETED', 'HKCU\\Software\\Microsoft\\Windows\\CurrentVersion\\Run', 'MicrosoftEdgeAutoLaunch_E27E50BD5633453D54C19716A813DFE6',
'"C:\\Program Files (x86)\\Microsoft\\Edge\\Application\\msedge.exe" --no-startup-window --win-session-start', None)
('ADDED', 'HKCU\\Software\\Microsoft\\Windows\\CurrentVersion\\Run', 'malware_test', None, 'C:\\Users\\Public\\malware.exe')
('ADDED', 'HKCU\\Software\\Microsoft\\Windows\\CurrentVersion\\Run', 'malware_test', None, 'C:\\Users\\Public\\malware.exe')
('DELETED', 'HKCU\\Software\\Microsoft\\Windows\\CurrentVersion\\Run', 'MicrosoftEdgeAutoLaunch_E27E50BD5633453D54C19716A813DFE6',
'"C:\\Program Files (x86)\\Microsoft\\Edge\\Application\\msedge.exe" --no-startup-window --win-session-start', None)
('ADDED', 'HKCU\\Software\\Microsoft\\Windows\\CurrentVersion\\Run', 'malware_test', None, 'C:\\Users\\Public\\malware.exe')
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PS C:\WindowsRegistryMonitor>
```

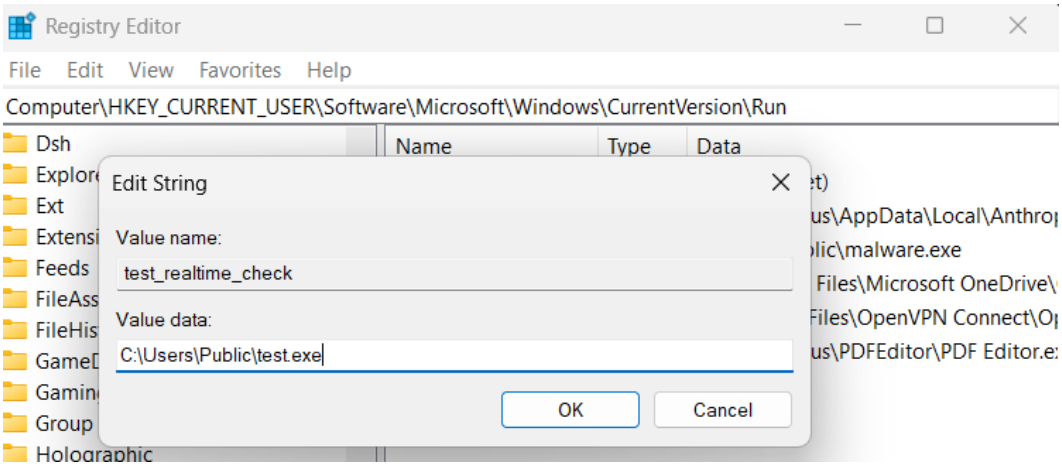

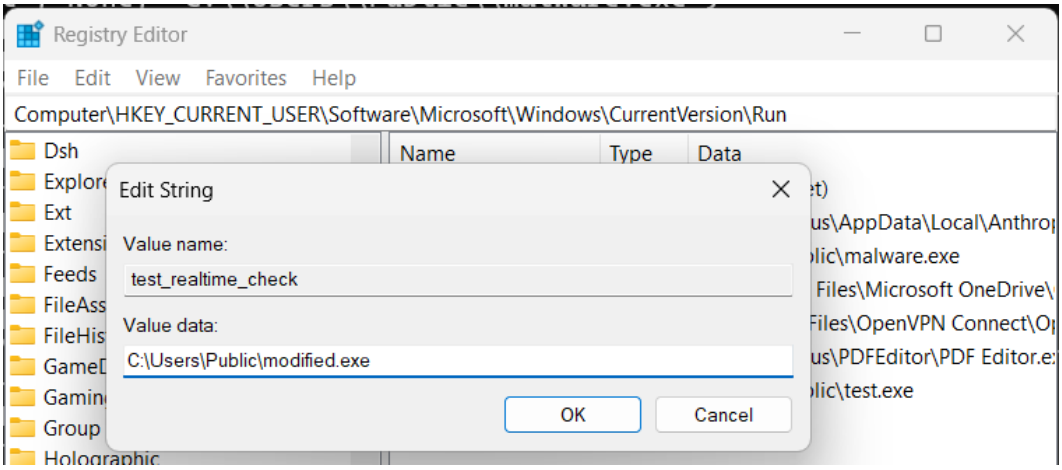


```
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PS C:\WindowsRegistryMonitor>
```

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```

```
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PS C:\WindowsRegistryMonitor>
```

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('ADDED', 'HKCU\\Software\\Microsoft\\Windows\\CurrentVersion\\Run', 'malwar
e_test', None, '')
PS C:\WindowsRegistryMonitor>
```

5. Explanation of Output

The console output displays detected changes in tuple format:
('CHANGE_TYPE', 'Registry_Path', 'Value_Name', 'Old_Value', 'New_Value')

Examples observed during testing:
- ADDED → New autorun entry detected.
- MODIFIED → Registry value altered.
- DELETED → Registry value removed.

These outputs confirm that the system successfully performs real-time registry monitoring.

6. Conclusion of Testing

The Windows Registry Monitoring System successfully detected unauthorized registry modifications in real time using polling-based baseline comparison. All changes were accurately captured and displayed in the monitoring console.